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Experience

Founding Robotics Software Engineer Tetheria.AI

Sept 2025 - Present

- Architected full software stack for 16-DOF and 20-DOF tendon-driven hands across firmware, SDK, and ROS2, establishing standardized coordinate frames, data structures, and control interfaces.
- Built and maintain Python SDK implementing low-level serial communication and binary command protocol at 500 Hz; guided firmware integration on ESP32 microcontrollers (350+ stars, 50+ forks).
- Designed and deployed neural network-based hand retargeting algorithm operating at 1000 Hz to map human hand motion to robotic hand kinematics (company IP).
- Developed control interfaces for 4+ robotic arm platforms and implemented custom inverse kinematics solvers with decoupled translation/rotation to mitigate singularities and motion jerk.
- Built teleoperation systems using VR trackers, Apple Vision Pro, MANUS Meta Gloves, and direct joint-mapping interfaces, enabling single-operator control of dual robotic arms and dual 16-DOF dexterous hands; deployed as full-stack bimanual data collection system.
- Led data collection, processing, and training pipeline for vision-language-action (VLA) models (e.g., Pi0.6, GR00T-N1.6), mentoring 2 engineers; deployed policies on physical hardware.
- Led technical interviews for robotics and AI engineering hires, influencing hiring decisions for 3 engineers.

Robotics Software Engineering Intern Tetheria.AI

May 2025 - Aug 2025

- Developed URDF from SolidWorks models and established consistent coordinate frame conventions; configured ROS 2 visualization stack including RViz2, TF alignment, launch files, and joint state publishing.
- Designed nonlinear joint-to-actuator modeling for tendon-driven hands, deriving motor commands from joint space using custom kinematic formulations.
- Implemented real-time hand teleoperation pipeline using motion capture gloves; returned full-time to lead continued platform development.

Research Assistant Robotics: Perception and Manipulation Lab, University of Minnesota

Jan 2024 - May 2025

- Implemented and deployed diffusion-policy based bimanual manipulation on real robot hardware (UR5e), achieving 74.6% task success on bottle uncorking task. (GitHub) (YouTube)
- Designed and deployed dual-arm manipulation pipelines on UR5e robots, integrating perception, motion planning, and collision avoidance (93.1% reduction in emergency stops). (YouTube)

Teaching Assistant University of Minnesota

Jan 2024 - May 2025

- Teaching Assistant across three schools (CS, Aerospace, Management), mentored 280+ students in robotics, dynamics, and applied data analytics; achieved 5.57/6 mean rating in Intelligent Robotic Systems course.

Scientific Officer NPCIL, India

Jan 2021 - Aug 2023

- Contributed to design of nuclear fuel handling machinery, performing structural and load calculations and supporting detailed engineering documentation; reduced design cycle time by 2+ months.

Projects

In-Context Imitation Learning via Graph Diffusion for Bimanual Manipulation (GitHub) (YouTube)

- Extended Instant Policy to dual-arm manipulation by designing heterogeneous graph-transformer diffusion architecture modeling grippers and scene jointly; trained 211M parameter model using programmatically generated pseudo-demonstrations.
- Achieved 74% success rate on RLBench2 Lift Tray task, outperforming ACT, RVT-LF, PerAct-LF, and PerAct2.

Education

University of Minnesota, MS in Robotics, GPA: 4.0/4.0

2025

Indian Institute of Technology (IIT) BHU Varanasi, B.Tech. in Mechanical Engineering, GPA: 7.2/10.0

2018

Skills

Programming: Python, C/C++, CUDA

Robot Learning & Control: ROS 2, Motion Planning, Generative Modeling, Graph Neural Networks, PyTorch

Systems & Infrastructure: Linux, Docker, Git, Embedded Control (ESP32), Serial Communication, AWS, Google Cloud